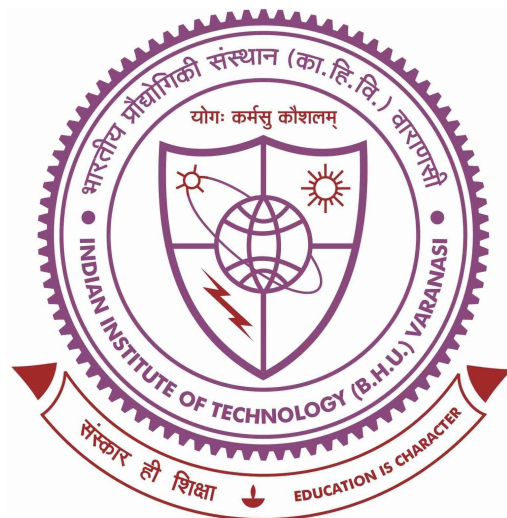


Towards phase-field modelling of phase inversion



Thesis submitted in partial fulfilment

for the Award of

INTEGRATED DUAL DEGREE (B.TECH.+ M.TECH.)

in

INDUSTRIAL CHEMISTRY

by

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Preface

This thesis presents a computational study on precipitate growth and microstructural evolution using phase-field modeling techniques. The work mainly focuses on understanding the effects of elastic interactions and interface curvature on precipitate growth through coupled Allen–Cahn and Cahn–Hilliard equations.

Phase transformations and microstructure evolution play an important role in determining the mechanical and physical properties of materials. In recent years, computational methods have become powerful tools for studying such phenomena. The present work aims to investigate these effects using numerical simulations and elastic energy formulations within the phase-field framework.

The research work presented in this thesis was carried out during my Integrated Dual Degree (B.Tech.+M.Tech.) program in the Department of Metallurgical Engineering. The study provided me with valuable experience in computational modeling, numerical methods, and scientific research.

I hope that the work presented in this thesis contributes toward a better understanding of elastic effects in precipitate evolution and phase transformations.

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Chapter 1

Introduction

1.1 Introduction

Microstructural evolution plays a crucial role in determining the mechanical, thermal, and physical properties of engineering materials. Processes such as phase separation, precipitate growth, coarsening, and solid state transformations strongly influence material behavior in alloys, ceramics, semiconductors, and functional materials. Understanding these processes is therefore important for the design and optimization of advanced materials systems.

During solid state phase transformations, the evolution of precipitates within a matrix phase is governed by thermodynamic and kinetic factors. In many alloy systems, precipitate growth occurs through diffusion-controlled mechanisms, where atoms diffuse from a supersaturated matrix into the precipitate phase. The morphology, growth rate, and spatial distribution of precipitates significantly affect material properties such as hardness, strength, creep resistance, conductivity, and phase stability.

In addition to chemical free energy, elastic interactions also play an important role in precipitate evolution. When the lattice parameters of the precipitate and matrix phases differ, elastic strain energy develops due to lattice mismatch. This misfit strain alters the local stress field and influences precipitate growth kinetics, morphology evolution, and

microstructural stability.

Traditional sharp-interface approaches, such as the Zener–Frank theory and the Laraia–Johnson–Voorhees (LJV) theory, have been widely used to study diffusion-controlled precipitate growth. However, these approaches require explicit interface tracking and become difficult to apply for complex evolving morphologies. To overcome these limitations, diffuse-interface approaches such as the phase-field method have become increasingly important.

1.2 Phase-Field Method

The phase-field method is a powerful computational technique used to model microstructural evolution without explicitly tracking interfaces. In this method, interfaces are represented as diffuse regions across which field variables vary continuously. The microstructure is described using continuous order parameters and composition fields.

The major advantage of the phase-field approach is its ability to naturally handle:

- complex interface evolution,
- precipitate coarsening,
- elastic interactions,
- morphology changes,
- and multiphase systems.

The phase-field framework is thermodynamically consistent and is based on the minimization of the total free energy of the system.

For systems involving conserved composition fields, the evolution is governed by the Cahn–Hilliard equation:

$$\frac{\partial c}{\partial t} = \nabla \cdot (M \nabla \mu) \quad (1.1)$$

where:

- c is the composition field,
- M is the mobility,
- μ is the chemical potential.

For non-conserved order parameters, the evolution is governed by the Allen–Cahn equation:

$$\frac{\partial \eta}{\partial t} = -L \frac{\delta F}{\delta \eta} \quad (1.2)$$

where:

- η is the order parameter,
- L is the relaxation coefficient,
- F is the total free energy.

The coupled solution of these equations enables simulation of precipitation, phase separation, and interface evolution.

1.3 Elastic Effects in Phase Transformations

Elastic stresses arise during phase transformations because of lattice mismatch between the matrix and precipitate phases. These stresses generate elastic strain energy, which contributes to the total free energy of the system.

The elastic contribution becomes especially important in coherent precipitates, where lattice continuity exists across the interface. The resulting elastic interactions can:

- modify precipitate growth rates,
- influence particle morphology,
- cause directional coarsening,
- and induce rafting phenomena.

The elastic energy is generally expressed as:

$$F^{el} = \frac{1}{2} \int_{\Omega} \sigma_{ij}^{el} \epsilon_{ij}^{el} d\Omega \quad (1.3)$$

where:

- σ_{ij}^{el} is the elastic stress tensor,
- ϵ_{ij}^{el} is the elastic strain tensor.

The elastic strain is obtained from:

$$\epsilon_{ij}^{el} = \epsilon_{ij} - \epsilon_{ij}^0 \quad (1.4)$$

where:

- ϵ_{ij} is the total strain,
- ϵ_{ij}^0 is the eigenstrain or misfit strain.

Mechanical equilibrium in the system is governed by:

$$\frac{\partial \sigma_{ij}}{\partial x_j} = 0 \quad (1.5)$$

Accurate computation of elastic fields is therefore essential for realistic microstructural simulations.

1.4 Numerical Methods for Elastic Field Calculation

Several numerical techniques have been developed for solving elastic equilibrium equations in phase-field models. Fourier spectral methods based on Fast Fourier Transform (FFT) are widely used because of their computational efficiency in periodic systems. These methods transform differential equations into algebraic equations in Fourier space and are particularly effective for periodic boundary conditions.

However, FFT-based spectral methods have limitations when non-periodic boundary conditions are required. Many real physical systems involve fixed, open, or non-periodic boundaries where spectral periodicity assumptions are not valid.

To address this limitation, the present work employs the Finite Difference Method (FDM) for solving the governing equations. In the FDM approach, spatial derivatives are approximated using finite difference approximations on discrete grids. This method provides flexibility in handling non-periodic boundary conditions and allows direct implementation of physically realistic boundary constraints.

Unlike FFT-based approaches, the present work solves the elastic equilibrium equations using iterative finite-difference discretization techniques under non-periodic boundary conditions.

1.5 Motivation of the Present Work

Most previous phase-field studies of precipitate growth involving elastic interactions have primarily used Fourier spectral methods with periodic boundary conditions. Although computationally efficient, these approaches may introduce artificial periodic interactions and may not accurately represent finite systems or systems with realistic physical boundaries.

The present work is motivated by the need to study precipitate growth using:

- non-periodic boundary conditions,
- finite difference discretization,
- and direct numerical solution of elastic equilibrium equations.

The implementation of finite difference techniques enables greater flexibility in modeling systems where periodicity assumptions are not appropriate.

In addition, the coupled solution of Allen–Cahn and Cahn–Hilliard equations with elastic energy contributions provides a comprehensive framework for studying diffusion-controlled

precipitate evolution.

1.6 Objectives of the Present Work

The main objectives of the present thesis are:

1. To develop a phase-field model for precipitate growth in elastically stressed systems.
2. To couple the Allen–Cahn and Cahn–Hilliard equations for microstructural evolution.
3. To incorporate elastic strain energy arising from lattice mismatch between precipitate and matrix phases.
4. To solve the elastic equilibrium equations using the Finite Difference Method (FDM).
5. To implement non-periodic boundary conditions in the numerical simulations.
6. To investigate the effect of elastic interactions on precipitate growth and morphology evolution.
7. To compare the simulated microstructural evolution with classical theoretical predictions and previous computational studies.

Chapter 2

Literature Review

2.1 Introduction

Microstructural evolution in multiphase systems has been extensively studied because of its influence on material properties. Earlier studies mainly focused on precipitate growth, coarsening, and elastic interactions during phase transformations. With the development of computational techniques, phase-field modelling became an important tool for studying such phenomena.

2.2 Classical Theory of Precipitate Growth

Zener and Frank developed the classical diffusion-controlled theory for precipitate growth in supersaturated alloys. Their model established the parabolic growth behavior of precipitates under diffusion-controlled conditions.

Later, Laraia, Johnson, and Voorhees incorporated coherent elastic strain energy into precipitate growth theory. Their work showed that elastic strain energy modifies precipitate growth kinetics and equilibrium conditions.

Although these theories provided important understanding of precipitate growth, they were based on sharp-interface assumptions and became difficult to apply for complex

evolving microstructures.

2.3 Development of Phase-Field Modelling

The phase-field method was developed to study microstructural evolution without explicit interface tracking.

Cahn and Hilliard introduced a diffuse-interface model for conserved composition evolution through the Cahn–Hilliard equation:

$$\frac{\partial c}{\partial t} = \nabla \cdot (M \nabla \mu) \quad (2.1)$$

Allen and Cahn later introduced an evolution equation for non-conserved order parameters:

$$\frac{\partial \eta}{\partial t} = -L \frac{\delta F}{\delta \eta} \quad (2.2)$$

These equations became the foundation of modern phase-field modelling and have been widely used for studying:

- phase separation,
- precipitate growth,
- coarsening,
- and morphology evolution.

2.4 Elastic Effects in Phase-Field Models

Khachaturyan developed the microelasticity theory for incorporating elastic interactions into heterogeneous systems. This framework enabled the inclusion of elastic strain energy in phase-field simulations.

Subsequent studies demonstrated that elastic interactions strongly influence:

- precipitate morphology,
- particle alignment,
- directional coarsening,
- and microstructural stability.

Elastic energy in coherent systems is generally expressed as:

$$F^{el} = \frac{1}{2} \int_{\Omega} \sigma_{ij}^{el} \epsilon_{ij}^{el} d\Omega \quad (2.3)$$

The associated mechanical equilibrium condition is:

$$\frac{\partial \sigma_{ij}}{\partial x_j} = 0 \quad (2.4)$$

2.5 Rafting and Elastic Inhomogeneity

Several studies investigated rafting behavior in elastically stressed systems.

Schmidt and Gross studied the effect of elastic modulus mismatch and applied stress on precipitate rafting. Their work showed that the direction of rafting depends on:

- **Applied stress,**
- **Eigenstrain,**
- and **Elastic Inhomogeneity.**

Gururajan and Abinandanan developed a phase-field model for studying rafting in elastically inhomogeneous systems. Their simulations demonstrated that elastic modulus mismatch significantly affects precipitate morphology and directional coarsening.

These studies suggested that elastic interactions may also influence topology evolution and phase connectivity.

2.6 Numerical Methods

Different numerical approaches have been used in phase-field simulations, including:

- **Finite Difference Method (FDM),**
- **Finite Element Method (FEM),**
- **Fourier spectral methods.**

FFT-based spectral methods are widely used because of their computational efficiency and spectral accuracy in periodic systems. However, these methods inherently assume periodic boundary conditions.

Finite Difference Method provides greater flexibility for:

- **non-periodic boundary conditions,**
- **finite-domain simulations,**
- **Iterative numerical implementation.**

Because of these advantages, FDM is suitable for studying topology evolution and connectivity changes in finite systems.

2.7 Research Gap

Most previous studies focused primarily on:

- precipitate growth,
- elastic coarsening,
- and rafting phenomena.

Relatively limited work has been carried out on phase inversion using phase-field methods with elastic interactions.

Furthermore, most elastic phase-field studies employed FFT-based spectral techniques

together with periodic boundary conditions.

The present work attempts to study topology evolution towards phase inversion using:

- coupled Allen–Cahn and Cahn–Hilliard equations,
- elastic free energy formulation,
- Finite Difference Method,
- and non-periodic boundary conditions.

Chapter 3

Methodology

Begin writing your methodology here.

Chapter 4

Results and Discussion

Begin writing your results here.

Chapter 5

Conclusion

Begin writing your conclusion here.